

Trust Calibration as a Core Mechanism of Bidirectional Human-AI Alignment

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Abstract

As AI systems increasingly act as collaborators and decision-support tools, alignment is no longer only a technical problem but also an interactional one. Bidirectional alignment emerges through ongoing adaptation between humans and AI systems, making it critical to understand how users interpret, rely on, and respond to AI outputs over time. This position paper argues that trust calibration is the core human-layer mechanism enabling this process. Alignment succeeds when users' trust appropriately matches system reliability, uncertainty, and purpose; when trust is miscalibrated, even technically robust systems may be over-relied upon or rejected. Drawing on research in human-automation trust, explainability, and value-sensitive design, we position trust calibration as a bridge to design, evaluation, and governance-aware implementation. We conclude with design implications, evaluation metrics, and a research agenda for value-centered human-AI co-evolution.

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1 Introduction

AI systems increasingly support open-ended tasks in domains such as medicine, law, and creative work. Early alignment research in AI focused on specifying objectives and preventing unintended behaviors [2, 25], aligning models to human preferences through feedback [5], and designing systems that support appropriate human control and reliance [1]. In everyday use, however, alignment is experienced less at the level of model objectives than at the level of interaction. Whether an AI system ultimately supports or undermines human goals depends on how people choose to use it: whether they accept its outputs, scrutinize them, revise them, or override them. These behavioral choices mediate the practical effects of alignment. We argue that *trust calibration* governs this mediation as it shapes users' expectations of system competence and intent, and determines the degree of reliance they grant under uncertainty [16]. When trust exceeds demonstrated reliability, automation bias and error propagation follow; when trust falls short, beneficial capabilities

remain unused [22]. Thus, bidirectional alignment, where humans and AI adapt to one another over time, depends not only on model updates, but on the continuous calibration of trust that regulates human reliance and feedback. This paper develops the argument by unpacking how trust calibration and bidirectional alignment operate together in human-AI interaction. It then discusses design implications, evaluation metrics, and a future research agenda for translating trust calibration into actionable implementation. By examining how trust shapes reliance, adaptation, and feedback over time, this paper connects alignment theory to interaction design and proposes trust calibration as a core design and evaluation target.

2 Trust as the behavioral gatekeeper of alignment

Research on trust in digital environments has shown that trust is shaped by perceptions of *ability/competence*, *integrity*, and *benevolence*, and that these perceptions systematically predict adoption, reliance, and compliance across e-commerce, online information systems, and automated decision aids [6, 13, 16, 19, 20, 22, 24]. This body of work converges on a core insight: trust is not simply an attitude, but a behavioral regulator that determines how people act under uncertainty. Hoff and Bashir proposed a three-layer model of trust (dispositional, situational, and learned) [14] for human-automation literature, a core theme in AI. Dispositional trust captures stable individual propensity; situational trust reflects task context; and learned trust updates through observed system performance and feedback. This framework is especially valuable for bidirectional alignment because it explains *why* the same model can produce different reliance behaviors across users and contexts, and *how* trust shifts over time as humans and AI systems co-adapt.

Trust calibration is uniquely important here not only because it predicts reliance, but because it captures the human layer at which alignment becomes behaviorally real. Technical properties and interface features do not determine aligned use on their own; what matters is how users interpret system competence, intent, and uncertainty, and whether those interpretations lead to appropriate reliance. Recent work on algorithm aversion and automation bias makes this clear: users may reject systems that outperform humans after observing errors [8], or over-accept recommendations without adequate verification [10]. A model may therefore be technically robust and still be misused or disused if users miscalibrate trust. Therefore, trust calibration should be a core construct beyond design features alone: design matters because it shapes trust, and trust in turn shapes whether alignment is enacted in practice.

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3 Bidirectional alignment and the operational role of trust

Much AI alignment research treats alignment as unidirectional: models learn human goals and values. In interactive settings, however, alignment emerges through reciprocal adaptation. Users update expectations and reliance based on experience, while AI systems adapt through explicit feedback or implicit behavioral signals. Hoff and Bashir's distinction between initial and dynamic learned trust explained how baseline expectations evolve with observed performance [14]. Trust calibration regulates human and AI's bidirectional alignment activities. When trust matches system reliability, users rely when appropriate and intervene when necessary. When miscalibrated, through over-trust or under-trust, errors propagate, monitoring declines, or engagement is withdrawn, distorting the feedback systems receive and degrading alignment over time.

This dynamic also makes values actionable. Value-centered alignment emphasizes embedding fairness, accountability, and autonomy into AI systems, but values are enacted through users' trust judgements. Lee and See describe trust in terms of perceived competence, integrity, and benevolence [16], mapping onto performance, process, and purpose. Value Sensitive Design calls for integrating values into technical and interaction processes [11], yet unless users can recognize and assess these qualities, alignment remains symbolic. Empirical work shows that transparency and explanations can shift trust without improving outcomes [3, 15], highlighting that trust calibration mediates value perception and reliance. In this sense, trust calibration is the mechanism through which bidirectional alignment becomes operational at the interaction level: it regulates when users defer, when they verify, and when they intervene, thereby shaping both immediate outcomes and the quality of feedback available for longer-term adaptation. Thus, bidirectional alignment requires not only value embedding and model adaptation, but interaction mechanisms that sustain calibrated trust.

4 From concept to actionable practice

Trust calibration becomes actionable when it is translated from an abstract mechanism into interactional requirements. A first step is recognizing that users do not calibrate trust from system performance alone, but from how systems communicate capability, uncertainty, purpose, and limits. Prior work in MATCH makes this point explicit by showing that interface features such as explanation format, confidence expression, interaction pacing, and recourse options shape the heuristics through which users decide whether to rely on, verify, or challenge AI outputs [18]. Trust calibration becomes more concrete when these judgments are further grounded in the questions users naturally ask while interacting with AI systems. Liao et al.'s XAI Question Bank captures this stakeholder perspective by identifying the kinds of questions systems should make answerable in context, including what the system did, why it behaved this way, how certain it is, and what recourse is available [17]. Framed this way, trust calibration also provides a practical bridge from broad governance principles, such as transparency, contestability, accountability, and human oversight, to concrete design and validation targets [9]. This translation is especially important

in LLM-based systems, where outputs are probabilistic, generative, and context-dependent. Together, these perspectives motivate our following discussion of how trust calibration can guide design principles, evaluation metrics, and a future research agenda.

5 Designing for trust calibration

We propose three design principles grounded in both foundational trust research and recent HAI studies.

Transparency aligned with performance. Explanations, uncertainty cues, and confidence indicators can support users' assessment of system competence. However, empirical studies in the LLM era show that explanations can increase perceived credibility even when objective performance does not improve [3]. Similarly, Ribeiro et al.'s LIME framework showed that local explanations shape user trust judgements [23], yet explanation alone does not guarantee appropriate reliance. Generative systems can also produce fluent and plausible outputs that obscure uncertainty and limitations [4]. The design challenge, then, is not simply to provide more explanation, but to communicate when the system is competent, when it is fallible, and when additional verification is warranted [3, 18, 28]. Transparency mechanisms should therefore be aligned with actual system reliability and paired with uncertainty communication and human-AI interaction guidelines [1].

Adaptive feedback and contestability. Bidirectional alignment depends on feedback loops that allow users to shape system behaviour. In LLM-based systems, interaction is often iterative and dialogic: users refine prompts, critique outputs, and co-construct responses. Reinforcement learning from human feedback (RLHF) operationalizes this adaptation at the model level [5], while interface affordances such as editable outputs, critique buttons, and prompt refinement tools enable behavioral steering. Recent studies of generative co-creation show that users develop evolving mental models through iterative engagement, and that visible correction mechanisms increase perceived control and trust stability [27, 28]. Contestability should therefore be designed as answerability. Interfaces can support calibrated trust by helping users ask why a result was produced, inspect alternatives, understand what changed after feedback, and identify how to correct or override problematic outputs [17]. In this way, contestability becomes an interactional requirement rather than an abstract governance principle, keeping trust dynamic rather than passive acceptance.

Maintaining human agency and adjustable autonomy. LLM-based systems blur boundaries between suggestion and authority. Studies of AI-assisted writing and coding reveal that users may defer to fluent outputs even when unsure of correctness, highlighting renewed risks of automation bias in generative settings [28]. Adjustable autonomy mechanisms—such as requiring confirmation for high-risk actions, offering multiple alternative outputs, or supporting staged review—can mitigate over-reliance. Contemporary human-centred AI frameworks emphasize graduated control and shared decision-making rather than binary delegation [1, 26]. These mechanisms matter not only because they preserve user control, but because they communicate that the system is advisory rather than authoritative, helping users remain epistemically engaged while still benefiting from automation [18]. Autonomy should therefore

be designed as negotiable and context-sensitive, supporting users as active decision-makers rather than passive recipients of AI output.

6 Evaluating bidirectional alignment through calibration

Traditional AI evaluation focuses on accuracy, robustness, and technical alignment. To assess bidirectional alignment, we must also evaluate how effectively systems support trust calibration. We propose several metrics:

- **Calibration accuracy.** Measure whether subjective trust matches objective reliability. For example, ask users to rate their confidence in AI outputs and compare these ratings against actual performance. Alignment improves when trust and reliability are correlated [7].
- **Appropriate reliance.** Assess whether users rely on AI when it is correct and override it when it errs. Bansal et al. showed that explanations increased acceptance regardless of correctness [3]; evaluation should reward systems that support selective reliance.
- **Trust resilience.** Examine how trust recovers after observed failures. Grgić-Hlača and colleagues investigated how human perceptions of fairness vary with personal experience and demographics [12]. Similar methods could assess whether trust perceptions stabilize after errors when appropriate repair mechanisms are provided.
- **Longitudinal co-adaptation.** Study how trust and user behaviour evolve over weeks or months of interaction. Longitudinal studies are critical for understanding dynamic learned trust and for revealing gradual shifts toward over-reliance or scepticism.
- **Interrogability and recourse.** Assess whether users can meaningfully question, challenge, revise, or override AI outputs, and whether these mechanisms improve calibration rather than merely increasing interaction volume [17].

Evaluations should combine behavioral metrics with subjective scales and societal impact measures such as perceived fairness, inclusivity, and collective well-being. This perspective also helps translate high-level governance goals into implementable evaluation criteria. Principles such as transparency, contestability, accountability, and human oversight are often stated abstractly in AI policy and regulation; trust calibration offers a more operational lens for assessing whether these principles are enacted in use. For example, transparency can be evaluated in terms of whether users can accurately judge system reliability, contestability in terms of whether they can effectively question and correct outputs, and human oversight in terms of whether interfaces support selective rather than passive reliance [9, 18]. Ultimately, alignment should be judged not only by technical correctness but by whether humans and AI systems form resilient, value-aligned partnerships.

7 A research agenda for BiAlign

Trust calibration frames a research agenda across disciplines:

- (1) **Integrative trust-alignment frameworks.** Combine models of human trust formation [14, 16] with computational

alignment techniques [5] to understand how system design influences trust dynamics.

- (2) **Longitudinal co-evolution studies.** Conduct multiweek experiments where humans interact with adaptive AI systems, tracking how trust and reliance evolve and how AI adapts to user feedback.
- (3) **Personalized calibration scaffolding.** Investigate how individual differences in culture and disposition influence trust, as evidenced by cross-cultural studies [12, 21], and design adaptive interfaces that tailor transparency and autonomy to individual preferences.
- (4) **Multi-level evaluation.** Integrate behavioral data, survey measures and societal indicators (e.g., fairness perceptions) to provide holistic assessments of alignment.
- (5) **Governance-to-design translation.** Study how broad requirements such as transparency, contestability, and human oversight can be translated into concrete interaction patterns, measurable calibration outcomes, and domain-specific validation practices [9].

Such an agenda requires collaboration between AI researchers, HCI designers, psychologists and social scientists and mirrors the interdisciplinary ethos of the human-AI bi-directional alignment.

Conclusion

This position paper argues that trust calibration is the core mechanism underlying bidirectional human-AI alignment. Alignment is enacted through interaction, and trust mediates whether AI systems are appropriately relied upon, challenged or ignored. Trust calibration operationalizes values, informs interface design and provides evaluation metrics. By foregrounding trust calibration, we move beyond static notions of alignment toward dynamic co-evolution between humans and AI systems. Advancing this agenda requires not only models that better reflect human values, but also interfaces that communicate system qualities responsibly, support stakeholder questioning, and translate governance principles into actionable forms of oversight and recourse. This perspective invites researchers and practitioners to develop systems that not only learn human values but also empower humans to calibrate trust, thus enabling resilient human-AI partnerships.

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